PMC

Testing Manual

v46p · Sovereign Edition

Per-module test cases · happy paths · edge cases · expected errors · HTTP codes

PMC MANAGERS · CONFIDENTIAL — INTERNAL USE ONLY

0 · How to use this manual

Every module has a fixed test-case format. Read the **Precondition** first — if you cannot satisfy it, either skip or arrange test data before continuing. Then follow **Steps** in order; do not paraphrase. Compare the actual UI against **Expected UI** and the network log against **Expected API**. Record any deviation on the tracker with the test-case ID.

Credentials live in `/app/memory/test_credentials.md`. Admin endpoints require **both** `Authorization: Bearer <token>` **and** `X-CSRF-Token: <csrf_token>`. The login response returns both.

Every AI-touched surface is advisory only; a failed AI call must never block a decision. If the AI proxy times out, the human path is always live.

Global error codes

Code	Meaning	Action
400	Bad request — payload validation failed	Check the field-level error map returned in the body.
401	Missing / expired token	Log in again. If persistent, clear cookies.
403	Wrong role	Check the require_role gate on the endpoint. Impersonate the right role.
404	Entity not found	Confirm the ref/id exists.
409	Conflict — envelope locked, duplicate resource, race	Refresh state; do not force-retry.
413	Payload too large — DPR > 200 MB	Split and use analyze-async endpoint.
422	Model validation error	Check request schema in the OpenAPI spec.
429	Rate-limited	Wait per Retry-After header.
502 / 504	Upstream (LLM, DocuSign, Stripe) hiccup	Retry with exponential backoff. Sev-3 alert if persistent.

CONFIDENTIAL — INTERNAL USE

1 · Module index

Public surfaces

- A · Public Notice Board

Vendor onboarding

- B · Vendor Registration

Auth

- C · 5-Portal Login

Admin — content

- D · Admin Console — Notices tab
- E · Admin Console — Tender Builder

Admin — decision

- F · Admin Console — Evaluation

Admin — money

- G · Admin Console — Escrow

Admin — auctions

- H · Admin Console — Reverse Auctions

Vendor daily

- I · Vendor Portal

Reporting

- J · Reports & Snapshot

PWA (v46p)

- K · PWA (v46p)

DENTIAL — INTERNAL USE

2 · Module — A · Public Notice Board

No auth required — anyone on the internet can browse open tenders.

A-01 · Landing page renders with 3 KPI cards

Precondition. None. Open incognito.

Steps

1. Navigate to /
2. Wait for hero + stats cards to load

Expected UI

- Hero shows 'Global Tender Board' headline, gold kicker, gold CTA.
- Three StatCards render: notices published, suppliers approved, packages total.
- 5-portal login CTA visible in header + hero as 'Portal gateways'.

Expected API

- `GET /api/v1/meta/public/stats → 200 JSON`
- `GET /api/v1/notices?limit=... → 200 items[]`

Test data / notes. Preview URL from frontend .env REACT_APP_BACKEND_URL

DENTIAL SE

A-02 · Filter notices by document type

Precondition. Landing loaded.

Steps

1. Click 'Tender' filter chip
2. Click 'RFP' filter chip
3. Click 'All' to reset

Expected UI

- Notice list re-renders instantly, only rows with matching document_type shown.
- Selected chip becomes solid navy; others revert to white/navy border.

Expected API

- `GET /api/v1/notices?document_type=tender → 200`

Test data / notes. Any published notice

A-03 · Filter by country

Precondition. Landing loaded, ≥ 1 notice.

Steps

1. Open country dropdown
2. Pick 'KZ' (Kazakhstan)

Expected UI

- List filters to KZ notices only. Dropdown label updates.

Expected API

- `GET /api/v1/notices?country=KZ → 200`

A-04 · Notice detail page opens

Precondition. ≥ 1 published notice.

Steps

1. Click any notice row card

Expected UI

- Route to /notices/{notice_ref}
- Detail renders: title, category, budget, deadline countdown, description, attachments panel.
- 'Days left' badge red when ≤ 14, otherwise navy.

Expected API

- `GET /api/v1/notices/{ref} → 200`

Test data / notes. PMC-NOT-2026-0001 or equivalent

A-05 · Q&A section on detail page

Precondition. Notice detail open, admin has published a Q&A.

Steps

1. Scroll to Q&A section
2. Expand any published question

Expected UI

- Q&A rows show question text + admin answer + published_at ISO date.

Expected API

- `GET /api/v1/notices/{ref}/qa → 200 items[]`

A-06 · About Us / Vision pages

Precondition. Public site.

Steps

1. Click 'About us' in top nav
2. Click 'Vision' in top nav

Expected UI

- Both pages render with sovereign navy hero + editorial content in EN/RU/KK per locale.

Expected API

- `No auth calls.`

DENTIAL — INTERNAL USE

3 · Module — B · Vendor Registration

Multi-step wizard with magic-link resume. Bug-blitz gate (v46p) enforces Section A completion before advancing.

B-01 · Start registration

Precondition. None.

Steps

1. Go to /register
2. Enter email 'qatest+{{uuid}}@example.com'
3. Optional: company name
4. Pick country KZ
5. Submit

Expected UI

- Toast success 'Draft created'
- Auto-redirect to /register/resume?token=<mMQ4...>

Expected API

- `POST /api/v1/registrations → 200 {id, magic_token}`

Test data / notes. Random email each run to avoid duplicate.

CONFIDENTIAL

B-02 · Section A gate — cannot jump to step B while incomplete

Precondition. Draft created (B-01).

Steps

1. On /register/resume, click stepper button 'B'
2. Observe toast + no navigation

Expected UI

- Step B/C/D/directors/E buttons show aria-disabled + opacity-40 cursor-not-allowed.
- Clicking any raises sonner toast 'Please fill Identity & Basic Info first. Missing: <fields>'.
- Section-a-gaps counter renders red under Next.

Expected API

- `No API call made.`

Test data / notes. Bug-blitz (a) verification — v46p.

B-03 · Complete Section A → gate opens

Precondition. Draft created.

Steps

1. Fill company_name, registered_address, country, registration_number, legal_entity_type
2. Click Save draft
3. Click stepper 'B'

Expected UI

- Stepper buttons become clickable; navigation advances to Section B.

Expected API

- `PATCH /api/v1/registrations/{id}?token=<t> → 200`

B-04 · Submit final requires directors + all declarations

Precondition. Sections A–D complete.

Steps

1. Click 'Submit final' with no director row

Expected UI

- Toast error 'At least one director / beneficial owner required'.
- directors-gap red hint visible.

Expected API

- `No POST /register/submit until fixed.`

B-05 · Magic-link resume from email

Precondition. Draft exists.

Steps

1. Open /register/resume?token=<stored_token>

Expected UI

- Draft state fully restored (all fields, stepper resumes at correct step).

Expected API

- `GET /api/v1/registrations/resume?token=<t> → 200`

B-06 · Duplicate email rejected

Precondition. Vendor already active with same email.

Steps

1. Go to /register, enter existing email, submit

Expected UI

- Toast error 'A vendor with this email already exists'. Form stays populated.

Expected API

- `POST /api/v1/registrations → 409`

4 · Module — C · 5-Portal Login

Sovereign gateway picker (Admin, Management, Partners, Agents, Vendors).

C-01 · Portal grid renders

Precondition. None.

Steps

1. Go to /login

Expected UI

- Five gradient portal cards visible.
- Each has kicker, name, description, ENTER PORTAL CTA.

Expected API

- `No API calls.`

C-02 · Admin credential flow

Precondition. Portal grid.

Steps

1. Click 'System Administrator' card
2. Login with adminsupport@pmcmanagers.com / Admin@12345

Expected UI

- Redirect to /admin/console.
- Response includes Bearer token + X-CSRF-Token (both required for subsequent admin endpoints).

Expected API

- `POST /api/v1/admin/auth/login → 200 {token, csrf_token}`

Test data / notes. test_credentials.md

C-03 · Vendor credential flow

Precondition. Portal grid.

Steps

1. Click 'Registered Vendors' card
2. Login with manualvendor@pmcmanagers.com / Vendor@12345

Expected UI

- Redirect to /vendor/dashboard.

Expected API

- `POST /api/v1/vendor/auth/login → 200`

C-04 · Wrong password path

Precondition. Portal grid.

Steps

1. Click any portal card, submit wrong password 3×

Expected UI

- Toast error 'Invalid credentials' on each attempt.
- After 5 attempts within 10 minutes, rate-limit banner appears.

Expected API

- `POST /login → 401 · repeated 401 → 429 after threshold`

5 · Module — D · Admin Console — Notices tab

Draft, publish, corrigenda, Q&A.

D-01 · Create a new notice from Tender Builder template

Precondition. Admin logged in. At least one master template exists.

Steps

1. Admin → Notices → New notice
2. Pick template PMC-KZ-CONSTR-001
3. Fill required fields (title, country, deadline, budget)
4. Click Publish now

Expected UI

- Modal closes. New row appears at top of Notices list with 'PUBLISHED' pill.
- Public /notices/{ref} shows the notice within ~30 s (CloudFront cache).

Expected API

- `POST /api/v1/notices → 201`
- `GET /api/v1/notices — new row present`

D-02 · EMD auto-compute honours slab

Precondition. New Notice modal open, tender value entered.

Steps

1. Click EMD Compute widget
2. Verify applied slab preview + computed EMD

Expected UI

- Applied slab shows band + percent + floor + ceiling.
- Computed EMD non-editable unless override reason provided.

Expected API

- `POST /api/v1/masters/emd/compute → 200 {computed, slab}`

D-03 · Publish a corrigendum

Precondition. Published notice exists.

Steps

1. Open notice → Corrigenda tab → New corrigendum
2. Enter reason + change body → Publish

Expected UI

- Corrigendum row appears with is_active=true.
- Vendors get an email via Resend within 2 min.

Expected API

- `POST /api/v1/corrigenda → 201`

D-04 · Withdraw a notice

Precondition. Published notice with no bids yet.

Steps

1. Open notice → 'Withdraw' action → confirm dialog

Expected UI

- Notice status flips to WITHDRAWN. Public detail returns 410.

Expected API

- `DELETE /api/v1/notices/{ref} → 200`
- `GET public → 410 gone`

D-05 · Q&A publish path

Precondition. Vendor has submitted a public question on a notice.

Steps

1. Admin → Notices → open notice → Q&A queue
2. Draft answer, mark 'Publish publicly', save

Expected UI

- Answer appears on public /notices/{ref} within 30 s.

Expected API

- `POST /api/v1/notices/{ref}/qa/answer → 200`

DENTIAL — INTERNAL USE

6 · Module — E · Admin Console — Tender Builder

Master + draft library, AI section drafter, DPR ingestion.

E-01 · Clone a master to a new draft

Precondition. Admin logged in, ≥ 1 master template.

Steps

1. Admin → Tender Builder → Master library
2. Click 'Clone as draft' on any master
3. Confirm new draft appears in 'Your drafts'

Expected UI

- New draft opens with prefilled sections. Save button initially disabled (no edits).

Expected API

- `POST /api/v1/tender-templates/clone → 201`

E-02 · AI per-section drafter (v46o hotfix)

Precondition. Draft open.

Steps

1. Click 'Build with AI' inside any section
2. Wait ≤ 20 s

Expected UI

- Section body auto-fills with 280-360 word HTML draft.
- Toast success 'AI draft generated'.

Expected API

- `POST /api/v1/tender-templates/{tid}/sections/{code}/ai-draft → 200 body[]`
- `Backend proxies via EMERGENT_LLM_KEY → integrations.emergentagent.com/llm`

Test data / notes. Regression check: prod key sk-emergent-*, no LLM_API_KEY.

E-03 · Async DPR ingestion (unlimited size · v46l)

Precondition. Draft open. Sample 100 MB PDF/DOCX.

Steps

1. Click 'Analyse DPR' → upload file
2. Poll status card until 'completed'

Expected UI

- Progress bar advances through phases: extract → chunk map → reduce → draft.
- On completion, every section body is populated from the DPR text.

Expected API

- `POST /api/v1/tender-pro/dpr/analyze-async → 202 {job_id}`
- `GET /api/v1/tender-pro/dpr/jobs/{job_id} → 200 {progress, phase, status}`

Test data / notes. Use any KMT DPR ≥ 500 pages.

E-04 · Save-as-master

Precondition. Draft with all sections filled.

Steps

1. Click 'Save as master' → confirm

Expected UI

- Draft copied into masters library. Original draft remains editable.

Expected API

- `POST /api/v1/tender-templates/{tid}/save-as-master → 201`

E-05 · Version snapshot + restore

Precondition. Draft with ≥ 3 edits.

Steps

1. Click History icon → pick prior snapshot → Restore

Expected UI

- Current draft saved as new snapshot, then reverted. Toast 'Restored older snapshot'.

Expected API

- `POST /api/v1/tender-templates/{tid}/versions/{v}/restore → 200`

DENTIAL — INTERNAL USE

7 · Module — F · Admin Console — Evaluation

Rubric, scoring, envelope locking, award report.

F-01 · Enter valid rubric

Precondition. Admin logged in. Notice with bids exists.

Steps

1. Admin → Evaluation → enter notice_ref
2. Adjust rubric so weights sum to 1.00
3. Click Save rubric

Expected UI

- Toast 'Rubric saved'. Save button re-enabled after modification.

Expected API

- `POST /api/v1/evaluations/rubric → 200`

F-02 · Rubric max gate (bug-blitz c)

Precondition. Rubric editor open.

Steps

1. Type commercial_weight = 1.5
2. Click Save (or press Enter programmatically)

Expected UI

- Save button disabled + red opacity-40.
- rubric-errors list shows 'Commercial weight must be between 0 and 1'.
- Even a keyboard/programmatic click fires the veto: no 'Rubric saved' toast, no POST.

Expected API

- `No POST /api/v1/evaluations/rubric`

Test data / notes. v46p defensive guard.

F-03 · Lock technical envelope

Precondition. Rubric saved, ≥ 1 technical score submitted.

Steps

1. Click 'Lock Technical'

Expected UI

- Button flips to 'Technical · Sealed ' with timestamp.
- Green banner records sealer + at-time.
- Further technical scoring rejected server-side with 409.

Expected API

- `POST /api/v1/evaluations/lock-envelope?kind=technical → 200`
- `Attempted score after lock → 409`

F-04 · Locked-eval reopen (bug-blitz d)

Precondition. Both envelopes locked on notice PMC-NOT-2026-0020.

Steps

1. Close panel, reopen, re-enter same notice_ref

Expected UI

- Envelope status banner shows sealed at + by.
- locked-eval-notice banner appears.
- AwardReport auto-loads.
- BidScoreList inputs disabled + Save Technical / Save Commercial disabled.

Expected API

- `GET /api/v1/evaluations/lock-state/{ref} → 200`
- `GET /api/v1/evaluations/award-report/{ref} → 200 {ranking, recommendation}`

Test data / notes. Verifies (d) fix.

F-05 · Generate award report

Precondition. Both envelopes locked, all scores in.

Steps

1. Click 'Generate report'

Expected UI

- Recommendation banner + ranking table.
- Any bid below minimum_technical shows Qualifies=—.

Expected API

- `POST /api/v1/evaluations/award-report/{ref} → 200`

F-06 · COI declaration

Precondition. Rubric saved.

Steps

1. Tick 'I have a conflict', enter details, click Declare COI

Expected UI

- Toast 'COI recorded'. Row appears in audit trail.

Expected API

- `POST /api/v1/evaluations/coi → 200`

8 · Module — G · Admin Console — Escrow

EMDs, joint accounts, transaction lifecycle.

G-01 · Create joint escrow account

Precondition. Admin logged in.

Steps

1. Admin → Escrow → New account
2. Enter PMC officer + vendor officer emails + name
3. Create

Expected UI

- Row appears with status 'pending'. Both officers get invite email via Resend.

Expected API

- `POST /api/v1/escrow/accounts → 201`

G-02 · Post EMD receipt

Precondition. Active joint account, bid submitted.

Steps

1. Escrow → New transaction → link to bid → enter amount, currency, receipt refs
2. Save

Expected UI

- Transaction row shows state 'held', aging 0 d.
- Vendor portal EMD tracker updates within 1 poll cycle.

Expected API

- `POST /api/v1/escrow/transactions → 201`

G-03 · Refund EMD

Precondition. EMD held, bid not awarded.

Steps

1. Open transaction → 'Initiate refund' → both officers sign
2. Refund confirmed on payment rail

Expected UI

- State transitions held → refund_pending → refunded.
- Aging clock resets. Audit row for each signature.

Expected API

- `PATCH /api/v1/escrow/transactions/{id}/refund → 200`

G-04 · Forfeit EMD (needs legal sign-off)

Precondition. EMD held, vendor withdrew after deadline.

Steps

1. Open transaction → 'Forfeit' → attach legal memo → both officers sign

Expected UI

- State transitions held → forfeited.
- Amount routes to sponsor holding account per Kazakhstan policy.

Expected API

- `PATCH /api/v1/escrow/transactions/{id}/forfeit → 200`

9 · Module — H · Admin Console — Reverse Auctions

Draft, start, close, invite, live bidding.

H-01 · Draft an auction

Precondition. Notice PMC-NOT-2026-XXXX exists with bids.

Steps

1. Admin → Live Auctions → New auction
2. Fill notice_ref, title, prices, decrement, duration, anti-snipe
3. Click 'Draft'

Expected UI

- Row created with status 'draft'. Bidder link hidden.

Expected API

- `POST /api/v1/reverse-auctions → 201`

H-02 · Start auction

Precondition. Draft exists.

Steps

1. Click 'Start' on the row

Expected UI

- Status flips to 'live', end_at populated, bidder URL exposed.
- Bulk-invite email fires to eligible bidders.

Expected API

- `POST /api/v1/reverse-auctions/{id}/start → 200`

H-03 · Close auction manually

Precondition. Auction is live, ≥ 1 valid bid.

Steps

1. Click 'Close' → confirm

Expected UI

- Winner banner shows alias + winning_price.
- State becomes 'closed'.

Expected API

- `POST /api/v1/reverse-auctions/{id}/close → 200 {winner}`

H-04 · View summary on mobile (bug-blitz e)

Precondition. Closed auction, viewport 390 × 844.

Steps

1. Open Admin → Live Auctions on iPhone SE emulator
2. Tap 'View summary' on closed row

Expected UI

- Toolbar wraps cleanly (flex-wrap gap-y-2).
- 'View summary' button fully inside card, no horizontal overflow.
- Summary panel renders below with start/floor/winning/savings columns.

Expected API

- `No new API call — data already in row.`

Test data / notes. v46p mobile fix.

10 · Module — I · Vendor Portal

Vendor-facing dashboard, KYC, licences, bids, EMDs.

I-01 · Vendor dashboard loads

Precondition. Logged in as manualvendor@pmcmanagers.com.

Steps

1. Navigate to /vendor/dashboard

Expected UI

- KPIs render: bids submitted, bids awarded, EMD held, next deadline.
- Trust-badge widget renders scorecard band.

Expected API

- `GET /api/v1/vendor/me → 200`
- `GET /api/v1/vendor/dashboard → 200`

I-02 · Submit a bid

Precondition. Logged in, open notice exists.

Steps

1. Vendor → Tenders → open notice → Bid
2. Fill price, upload technical + commercial docs, submit

Expected UI

- Bid row appears in vendor's My Bids with status 'submitted'.
- EMD requirement banner appears if not yet posted.

Expected API

- `POST /api/v1/bids → 201`

I-03 · Track bid status

Precondition. ≥ 1 submitted bid.

Steps

1. Vendor → My Bids

Expected UI

- Each bid row shows status timeline: submitted → under review → eval-locked → awarded/rejected.
- Awarded bid shows PO number + LOA link.

Expected API

- `GET /api/v1/vendor/bids → 200 items[]`

I-04 · Auction join (masked identity)

Precondition. Vendor invited to live auction.

Steps

1. Follow auction invite link → /vendor/auction/{id}

Expected UI

- Alias assigned (never real company name).
- Countdown clock live. Bid input allows only below current * (1 - decrement).

Expected API

- `POST /api/v1/reverse-auctions/{id}/register → 200 {alias}`

11 · Module — J · Reports & Snapshot

Stakeholder reports + on-demand snapshot to repo.

J-01 · Fetch stakeholder-report/tenders JSON

Precondition. Admin auth token in header.

Steps

1. curl -H 'Authorization: Bearer <t>' -H 'X-CSRF-Token: <c>' GET /api/v1/stakeholder-reports/tenders

Expected UI

- 200 JSON with generated_at + items[]. Each item has notice_ref, bids_total, active_corrigenda, days_since_publish.

Expected API

- `200`

J-02 · Fetch stakeholder-report/escrow CSV

Precondition. Same auth.

Steps

1. curl -X GET /api/v1/stakeholder-reports/escrow?format=csv

Expected UI

- Content-Type text/csv; two sections: --- accounts --- and --- transactions ---.

Expected API

- `200`

`J-03` · Save reports snapshot to disk (bug-blitz b)

Precondition. Same auth.

Steps

1. POST /api/v1/reports/snapshot

Expected UI

- Always returns valid JSON (ok, generated_at, files[], errors[]).
- Even on WeasyPrint failure the response parses cleanly — no HTML 500.

Expected API

- `200 JSON (never text/html)`

Test data / notes. v46p safety wrapper.

DENTIAL — INTERNA SE

12 · Module — K · PWA (v46p)

Installable on Android + iOS. Offline app shell.

K-01 · Manifest served

Precondition. Preview URL up.

Steps

1. curl -s /manifest.json | jq .

Expected UI

- JSON with name, short_name, icons (192 + 512), start_url, display=standalone, theme_color.

Expected API

- `200 application/json`

K-02 · iOS meta tags present

Precondition. None.

Steps

1. curl -s / | grep apple-mobile-web-app

Expected UI

- apple-mobile-web-app-capable=yes, status-bar-style=black-translucent, apple-touch-icon linked.

Expected API

- `200`

K-03 · Install on Android (Chrome)

Precondition. Chrome ≥ 90, HTTPS.

Steps

1. Open the URL, wait 30 s (engagement heuristic)
2. Chrome menu → 'Install app'

Expected UI

- App installs, launches in standalone window with PMC crest and navy splash.

Expected API

-

K-04 · Install on iOS (Safari 16.4+)

Precondition. iOS 16.4+.

Steps

1. Open URL in Safari → Share → 'Add to Home Screen'

Expected UI

- Icon on homescreen. Tap opens in standalone view; no Safari chrome.

Expected API

-

K-05 · Service worker caches app shell

Precondition. PWA installed.

Steps

1. Open DevTools → Application → Service Workers

Expected UI

- SW status = activated + running.
- Cache Storage lists 'pmc-shell-v46p' with HTML/CSS/JS and PMC crest.

Expected API

-

K-06 · API requests bypass cache

Precondition. Any online session.

Steps

1. Watch DevTools Network while browsing notices

Expected UI

- /api/* requests never marked 'from ServiceWorker'.
- Bid data always fresh from backend.

Expected API

-

K-07 · Offline shell shows

Precondition. PWA installed.

Steps

1. Toggle DevTools 'Offline'
2. Reload the page

Expected UI

- App shell renders. Data endpoints show reconnect banner.
- Once back online, banner auto-clears within 1 poll cycle.

Expected API

-

13 · Sign-off

This testing manual is versioned under `/app/deliverables/v46_manual/build/`. Regenerate with `python3 /app/tools/manual/testing_manual.py` whenever a module changes materially. Every test-case ID is stable — do not renumber; add new cases with a new letter suffix (e.g. F-04a).

DENTIAL — INTERNAL USE